



LESTER B. PEARSON
Pearson Electrotechnology Centre

LAB 4

PABX

Module 20

Connection to the Telephone Network

790495

Objective

The objective of this lab is to familiarize students with the basic configuration, operation, and feature programming of a small-scale PABX system, using the Excelltel MS208 model as a practical reference. Throughout this activity, students will learn how to access the programming mode, enable and disable C.O. lines, assign ringing extensions, record a welcome message (OGM), activate music-on-hold, and test call handling functions such as transfer and conference.

This lab simulates a real-world telecommunications environment where technicians or installers are required to configure a PABX for small offices or residential applications. By performing these procedures directly on the system, students develop both technical confidence and troubleshooting skills, preparing them for future field work involving telephony and communication infrastructure.

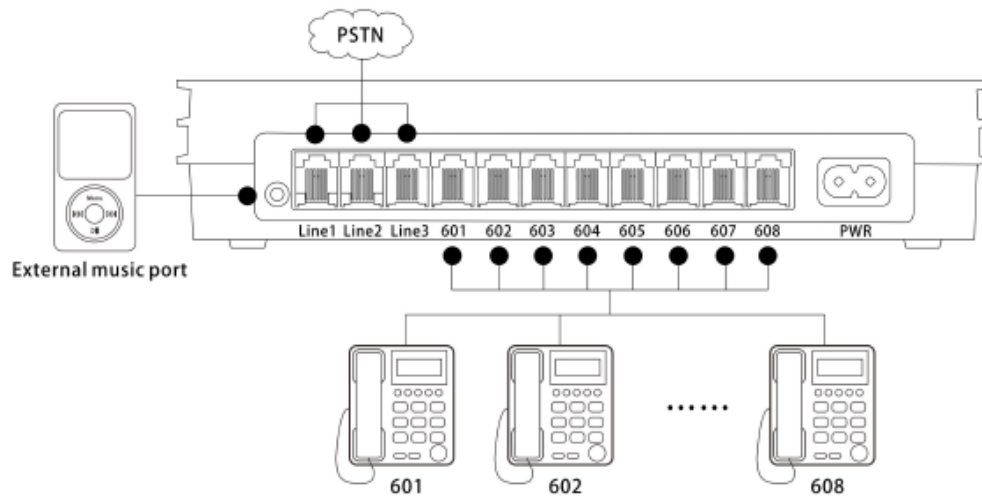
Scenario

Students will work in teams of two. Each student will use the telephone line installed at their assigned cubicle and connect it to the PABX system, occupying C.O. Line 1 and C.O. Line 2 on the Excelltel MS208. C.O. Line 3 must remain disabled for this activity, simulating an installation with limited service capacity.

All programming and configuration will be performed from extensions 601 and 602, as these are the designated programming extensions for the system. Each pair is responsible for verifying line activation, ringing behaviour, welcome message playback, and call handling functions using only the resources available at their workstation.

This setup simulates a small office or residential deployment where technicians must activate only the required lines, disable unused connections, and configure system behaviour according to customer needs.

208/308 Connection Diagram:



Part 1

Enter Programming Mode (*#*0000)

Reference image from manual:

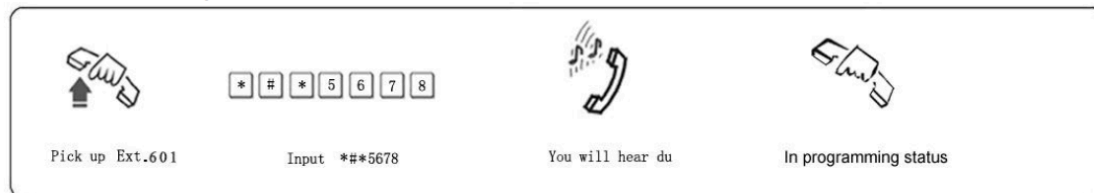
3.2 Basic Command

3.2.1 Enter into System Programming Status

Format: # * EFGH

Explanation: EFGH is the system password, default is 0000.

Example: To programming the system, you should input system password at first, if the password is 5678, as below:



- You need to dial: * # * EFGH in Direct-Dialing-Out mode.
- The default password is 0000, so you need to input *#* 0000 at the first time when operate this system.
- Programming Ext is EXT.601, or EXT.602, or EXT.603, or EXT. 604

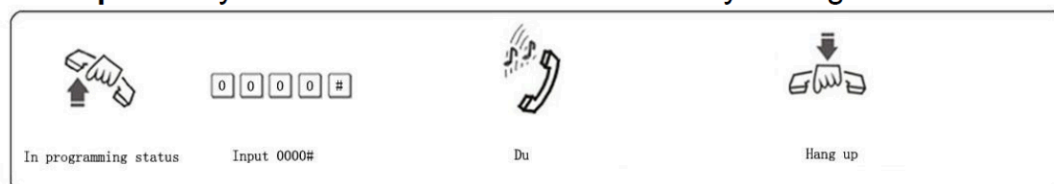
Part 2 - Reset to Default Configuration

Before beginning any programming tasks, it is essential to reset the PABX to its default factory settings. Because these units are reused by multiple student groups, we cannot assume what features, extensions, call routing behaviors, or line assignments may have been previously configured. Resetting the system ensures that every team starts from the same baseline and avoids configuration conflicts that would affect testing results.

3.2.4 System Restore Default Setting

Format: 0000 #

Description: System will restore the default factory setting.



- This command has no influence on the status of enabling/disabling C.O. line.

Part 3 - Enabling C.O. Lines

In a PABX system, the C.O. (Central Office) Lines represent the incoming telephone lines provided by the service source. Enabling or disabling C.O. lines determines which external lines are available for calling, receiving calls, and routing through the system.

In this lab, each student team is assigned one physical telephone line, and because teams work in pairs, the system will operate with a total of two active C.O. lines. For this reason, only C.O. Line 1 and C.O. Line 2 must be enabled. Any unused lines, such as C.O. Line 3, must remain disabled to prevent call routing issues, false line detection, or system misbehavior caused by open/unused ports.

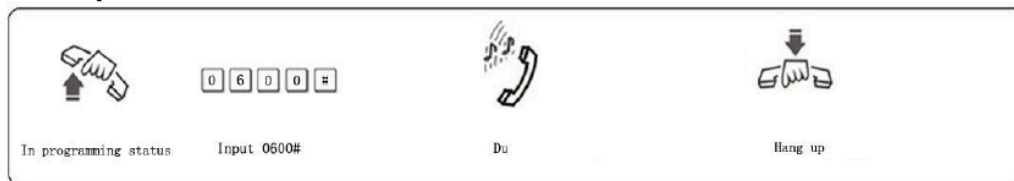
Enable Only C.O. Line 1 and Line 2 (0601#, 0602# / 0613#)

Reference image from manual:

3.3.1 Enable All C.O. Lines

Format: 0600#

Description: Enable all C.O. lines



 On default setting: system is in CO line disabled status.

3.3.2 Disable All C.O. Lines

Format: 0610#

Description: Disable all C.O. lines

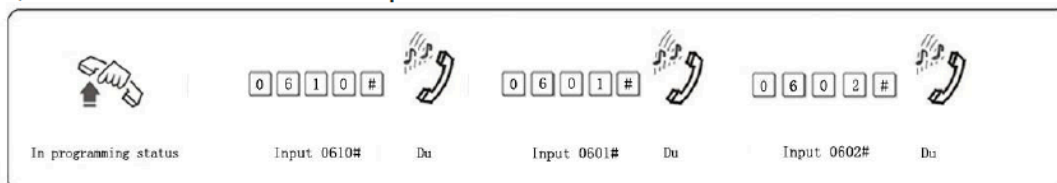
3.3.3 Enable No. M C.O. Line

Format: 060M#

Explanation: M: C.O. Line port No. M = 1 - 3

Description: Enable the No.M C.O. line

Example: You have a 308 PBX, you want to use C.O. line 1 and C.O. line 2, but the line 3 is not occupied.



3.3.4 Disable No.M C.O. Line

Format: 061M#

Explanation: M: C.O. Line port No. M = 1 - 3

Description: Disable the No.M C.O. line

Part 3 Assign Ring to Extension 601

3.8 Set Operator and Ringing Extension

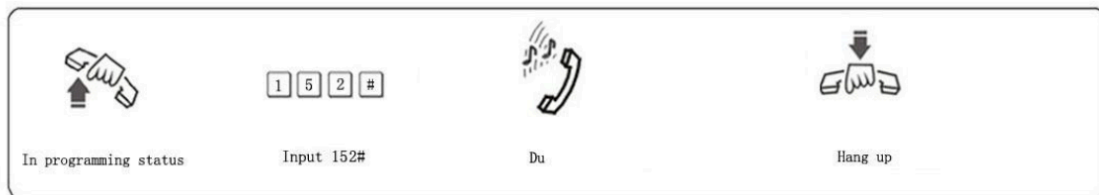
3.8.1 Set Operator

Format: 15N #

Explanation: N: the Ext. port No., N=1-8

Description: To assign any Ext as the operator. In the default setting, Ext 602 is the operator.

Example: Set Ext.602 as operator



Part 4 Direct Ring Assignment per Line

In this configuration, each C.O. line is assigned to ring a specific extension directly. According to the behavior described in section 3.8.2 of the manual, the MS208 allows individual routing so that each incoming line can ring a designated endpoint instead of following the default operator pattern. For this lab, the routing must be configured as follows:

C.O. Line 1 → Extension 601

C.O. Line 2 → Extension 602

This ensures that each student in the pair is responsible for their own incoming call path, supports controlled troubleshooting, and reflects real-world installations where each external line may serve a different department, desk, or role.

3.8.2 Set Ringing Extensions in Day mode

Format: 3 0 M N 0 #

Explanation:

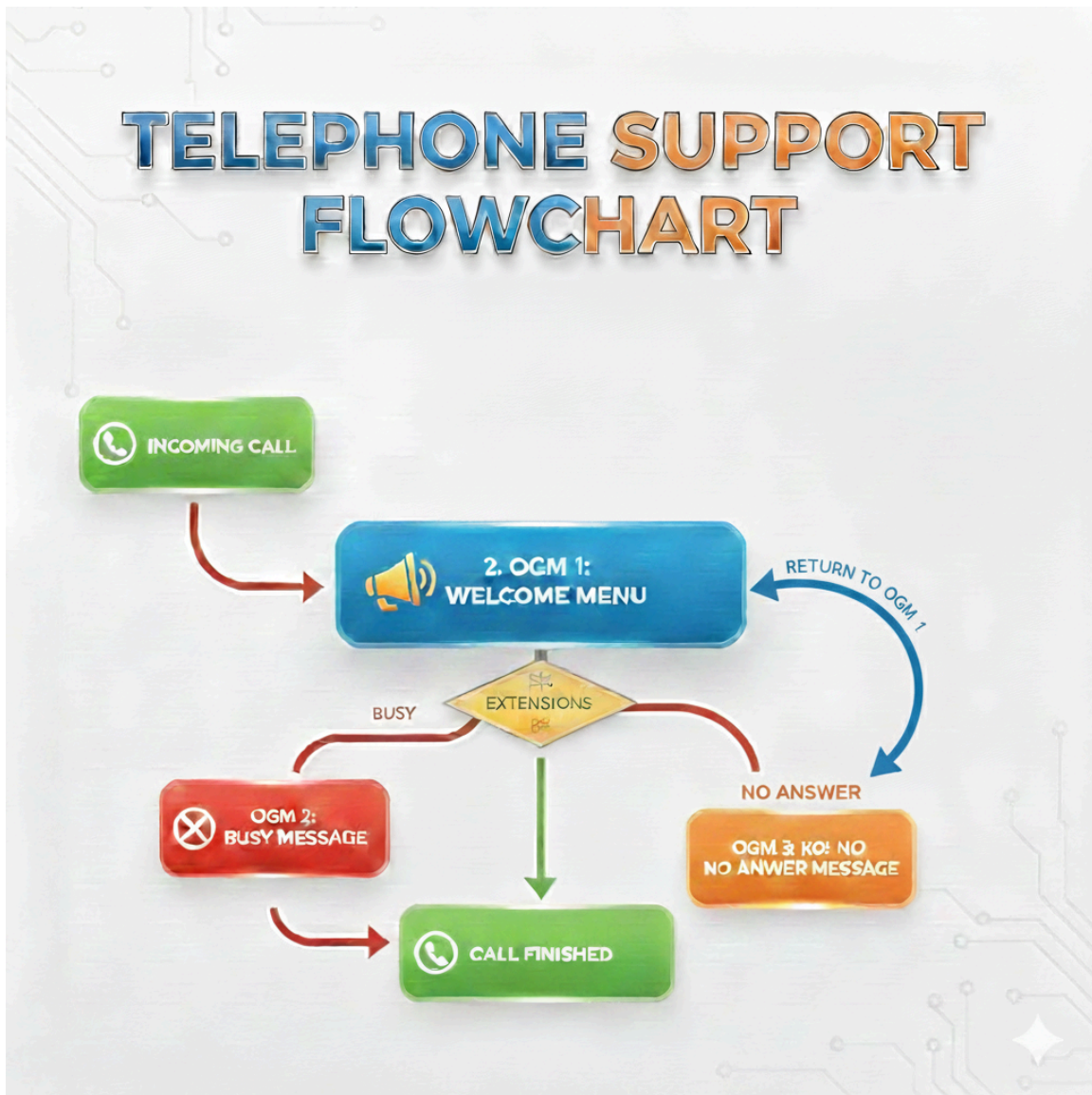
M: the C.O. Line port No. M=1 - 3

N: the Ext. port No. N=1–8(N=1 means Ext.601 N=8 means Ext.608)

Description: In operator work mode when incoming call from No.M C.O. line arrives and system in day mode(in default setting system in day mode), No. N Ext. will ring.

In default setting only Ext601, 602 will ring.

Welcome Message / OGM (Record: 12M#, Playback: 13M#)



Reference image from manual:

be set in four modes, see below:

Format: 0451# Set OGM in one section mode

0452# Set OGM in two section mode

Description: In two section mode, it is day/night mode, while in day time system use the first section, in night time system use the second section.

Format: 0453# Set OGM in three section mode;

(this is default setting)

0454# Set OGM in four section mode

Description: In four section mode, while in day-time the system use the first three section message, in night-time the system use the fourth section message)(**Note: this mode is only available when OGM recordable time is 60s or 90s**)

3.6.2 OGM recordable time

For 20 seconds type(standard type)

- ◆ In one section mode is total 20 seconds.
- ◆ In two section mode, each section is 10seconds
- ◆ In three section mode, the first section is 8 seconds, the second & the third section is 6 seconds

For 60 seconds type(optional for higher version MS+ series)

- ◆ In one section mode is total 60 seconds.
- ◆ In two section mode, each section is 30seconds
- ◆ In three section mode, each section is 20seconds
- ◆ In four section mode, the first & fourth section is 20seconds, the second & the third section is 10seconds

For 90 seconds type(optional for higher version MS+ series)

- ◆ In one section mode is total 90 seconds.
- ◆ In two section mode, each section is 45seconds
- ◆ In three section mode, each section is 30seconds
- ◆ In four section mode , the first & forth is 30 seconds , the second & the third section is 15 sections .

3.6.3 OGM Recording(multiple section mode)

Format: 1 2 M # (M=1,2,3, 4, the section of the OGM)

Description: to record an greeting message for Direct Inward System Access(DISA).

OGM 1 recording, this is the first OGM, usually it is for company greeting, for example, you may record "This is ABC company, please dial extension number or dial "0" for operator. Thanks for your call."

OGM 2 recording, this is the second OGM, usually it is for C.O. line dial extension and this extension is busy, if you want to transfer this calling to other extension, you may record "Sorry! This extension is busy. please dial other extension number or dial "0" for operator."

OGM 3 recording , this is the third OGM, usually it is for C.O. line dial extension and this extension no answer, you may record "Sorry! This

extension do not answer, please dial other extension number or dial "0" for operator."

- The recording time is limited, when you hear a beep tone "du" to start recording and another beep tone "du" to end recording;
- The new message will overwrite the old one;
- Please use high quality telephone to ensure high quality of recording;
- If the caller dial a number before the system finish playing the message, the system will terminate playing the message, waiting for the caller to continue with dialing;
- The call will be forwarded to Operator if caller dial 0.

3.6.4 OGM playing(multiple section mode)

Format: 13 M # (M=1,2,3, 4, the section of the OGM)

Description: Check and review the recorded message, if you are not satisfied, you can record again.

3.6.5 One section OGM Mode

Format: 1200 #

Description: to recording one section OGM

Format: 1300 #

Description: to play and check one section OGM

3.7 Review the Holding Music

Format: 14 #

Description: Perform the music on holding melody or test the quality of telephone unit.



3.8 Set Operator and Ringing Extension

3.8.1 Set Operator

Format: 15N #

Explanation: N: the Ext. port No., N=1-8

Description: To assign any Ext as the operator. In the default setting, Ext 602 is the operator.

Example: Set Ext.602 as operator

Music On Hold (MOH) – Enable: 14#

Reference image from manual:

extension do not answer, please dial other extension number or dial "0" for operator."

- The recording time is limited, when you hear a beep tone "du" to start recording and another beep tone "du" to end recording;
- The new message will overwrite the old one;
- Please use high quality telephone to ensure high quality of recording;
- If the caller dial a number before the system finish playing the message, the system will terminate playing the message, waiting for the caller to continue with dialing;
- The call will be forwarded to Operator if caller dial 0.

3.6.4 OGM playing(multiple section mode)

Format: 13 M # (M=1,2,3, 4, the section of the OGM)

Description: Check and review the recorded message, if you are not satisfied, you can record again.

3.6.5 One section OGM Mode

Format: 1200 #

Description: to recording one section OGM

Format: 1300 #

Description: to play and check one section OGM

3.7 Review the Holding Music

Format: 14 #

Description: Perform the music on holding melody or test the quality of telephone unit.



3.8 Set Operator and Ringing Extension

3.8.1 Set Operator

Format: 15N #

Explanation: N: the Ext. port No., N=1-8

Description: To assign any Ext as the operator. In the default setting, Ext 602 is the operator.

Example: Set Ext.602 as operator

Call Transfer (Flash + Extension)

Reference image from manual:

3.21 Ext. Group Setting

3.21.1 Set Ext. Group Number

All the extension can be set in 2 groups.

Format: 9 9 0 abcd #

Description: set the first group number and enable it.

Format: 9 9 1 abcd #

Description: set the second group number and enable it.

Use:

- The length of Ext group number should be the same with the first extension number

- Ext group number should be different with the other extension number

3.21.2 Add members in Ext. group

Format: 3 9 0 #

Description: to delete all extension members in the first group.

Format: 3 9 0 N₁ N₂ ... N_R * N_S .. N_T #

Explanation: N₁, N₂, N_R, N_S, N_T the Ext. port No(N= 1-8 means ext 601-608)

Description: In the first group join the Ext.N₁, Ext.N₂...Ext.N_T and from Ext. N_R to Ext.N_S is in sequence member.

Example: Add Ext.601, Ext.602, Ext.605 ...to Ext.608(include Ext.608) in sequence as members of the first group, then input 390 1 2 5 * 8 #.

After by this setting, when you dial the group number, Ext.601, Ext.602, Ext.605 ...Ext.608 will ringing at the same time.

Format: 3 9 1 #

Description: to delete all extension members in the second group.

Format: 3 9 1 N₁ N₂ ... N_R * N_S .. N_T #

Description: In the second group add Ext.N₁, Ext.N₂...Ext.N_T and from Ext.N_R to Ext.N_S is in sequence member.

3.21.3 Set the Ext. Group work mode

Format: 1 6 8 #

Description: Set the Ext. group in meeting mode.

Explanation: when Ext. group is in meeting mode; If some extension call this group, all the members in this group will ring simultaneously, each extension can pick up the handset to talk with the caller, while the other extension keep ringing. This is also a conference function.

Instruction: 1 6 9 #

Description: Set the Ext group in normal call mode.

Explanation: when Ext group is in normal call mode; If call this group, all the members in this group will ring simultaneously, each extension can pick up the handset to talk with the caller, if any extension answer the call, the other extension will stop ringing.

◆ Default setting is normal call mode.

Chap 4 Operating Instruction

4.1 Day/Night Work Mode Setting

4.1.1 Manual Mode Setting

In manual mode, you can input the following command:

Format: # 2 0# Set the system in day work mode.

2 1# Set the system in night work mode.



• On default setting, system is in day work mode.

• This setting can be only performed by programming extension (Ext.601, 602, 603, 604), no need in programming status.

• If Ext. is in direct-dialing-out mode, you need to dial *#20# or *#21#.

4.1.2 Automatic Mode Setting(Optional only for higher version MS+ series)

Format: # 2 2 # Set day/night in auto mode (Same as 040# in programming mode).

2 3# Set day/night in manual mode(Same as 041# in programming mode).

This setting can be only performed by programming extension (Ext.601, 602, 603, 604), no need to enter in programming status.

NOTE:

◆ There are two ways to switch Day/Night mode, Manual mode and Auto mode. Auto mode is optional function.

◆ In Manual mode, you need to switch Day/Night mode through setting manually, that means every day you must set manually if you want switch Day/Night mode. The system will be in Day mode (or Night mode)always if you don't set manually.

◆ In Auto mode, the system will switch between the day and night mode automatically according to the start and end time you set.

◆ In Auto mode, the system also will switch between the day and night mode automatically according to the weekend mode you set.

4.2 Set & Clear other Ext. Call forward through Programming Extensions

You can set other Ext. call forwarding through Programming Extensions

Format:

#26Nabcd: Set No.N Ext. always forward incoming call automatically to Ext.abcd

2 6 N #: Clear always forward setting for No.N Ext.

2 6 #: Clear always forward setting for all Ext.

2 7 N abcd: Set No.N Ext. forward incoming call to Ext.abcd when No.N Ext. is busy or no answer after 30 seconds(call forward time, refer to 3.19.3)

Conference Call (Flash + * + Extension)

Reference image from manual:

2 7 N #: Clear busy(or no answer)forward setting for No.N Ext.

2 7 #: Clear busy (or no answer) forward setting for all Ext.

2 5 #: Clear any call forward setting for all Ext no matter always forwarding setting or busy(no answer).

Explanation: "abcd" is the Ext number to be forwarded, abcd=601–608(default number) N: Ext. port No ,N=1-8 (N=1 means Ext.601... N=8 means Ext.608)

4.3 Outgoing Call

The system has 5 dial out mode

4.3.1 Outgoing Call Dial 9 at first (or Dial Directly)

Format: Pick up the handset and dial 9 at first after hear internal dial tone, then dial the desired number after hear the C.O line dial tone.

Description: When Ext is in indirect-Dialing-Out mode, you will hear internal dial tone after pick up the handset, dial 9 then you access the C.O. line, you can dial the number after hear C.O line dial tone.

Example: You want to call the external number 80129876 in indirect-Dialing-Out mode



- You need to dial 9 within 15 seconds after hear internal dial tone.
- If Ext. is in direct-Dialing-Out mode, you can pick up the handset and dial number directly after hear C.O. Line dial tone. In this situation you can redial the last dialed number.

4.3.2 Outgoing Call by Accessing C.O Line (except for 108)

Format: # 1 M (Indirect-Dialing-Out mode, refer chap 3).

Explanation: M: C.O. line port number(M = 1 -3)

Description: You can choose certain C.O line to make outgoing call.

Example: In indirect-Dialing-Out mode, you need to dial external number by accessing C.O. line3



- If the Ext is at Direct-Dialing-Out mode, you should input: * # 1M.

STEPS

1. Enter Programming Mode

*#*0000

2. Factory Reset

0000#

3. Disable all C.O. Lines

0610#

4. Enable C.O. Line 1

0601#

5. Enable C.O. Line 2

0602#

6. Set Operator Mode

11#

7. Clear ringing assignment C.O. Line 1

301#

8. Assign C.O. Line 1 → EXT 601

30110#

9. Clear ringing assignment C.O. Line 2

302#

10. Assign C.O. Line 2 → EXT 602

30220#

TEST:

Call C.O. Line 1 → 601 must ring

Call C.O. Line 2 → 602 must ring

11. Set One-Section OGM

0451#

12. Record OGM

1200#

13. Playback OGM

1300#

14. Activate Auto-Attendant

10#

TEST:

Call C.O. line → hear recorded OGM

15. Return to Operator Mode

11#

16. Test Music On Hold

14#

17. Call Transfer

FLASH + extension

18. Three-Way Conference

FLASH + * + extension